

Eduardo Alvarez

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EDUCATION

University of Nebraska-Lincoln

Lincoln, NE

Bachelor of Science in Mechanical Engineering

Expected Graduation May 2028

EXPERIENCE

Husker Motorsports Formula SAE Team

Lincoln, NE

Suspension/Vehicle Dynamics Director

August 2024-Present

- Led vehicle dynamics development by analyzing tire behavior, load transfer, suspension kinematics, and handling balance, resulting in a top team suspension setup.
- Analyzed Formula SAE Tire Test Consortium data in self-made MATLAB script to compare relationships of lateral force, slip angle, cornering stiffness, pressure, inclination angle, and vertical load, resulting in data driven decision for the best tire and setup.
- Directed the design and integration of the inboard and outboard suspension systems, including rockers, pushrods, control arms, uprights, hubs, and mounting components, resulting in a fully package suspension system with verified clearance.
- Designed a decoupled inboard suspension actuation system in SolidWorks; performed component level FEA to verify stress met required factors of safety and executed clearance checks across inboard packaging.
- Operated CNC mills and lathes, plus manual mill/lathe equipment, to manufacture high-quality components to print specifications and tight tolerance requirements.
- Maintained engineering documentation and revision control to ensure traceability of design decisions and change history.

Husker Hub, University of Nebraska-Lincoln

Lincoln, NE

Peer Specialist

August 2024-Present

- Guided students and families on FAFSA, scholarships, grants, loans, enrollment verification, and billing; clearly explained processes to ensure accurate, compliant financial aid information.
- Collaborated cross-functionally with university offices via Microsoft Teams to gather and share accurate information, improving students' access to reliable financial resources.
- Assisted 1,000+ students and families weekly with financial aid questions by providing clear, accurate guidance to support informed funding decisions.

Rigo's Landscape LLC

Olathe, KS

Landscaper

June 2013-August 2023

- Collaborated with cross-functional crews to deliver landscaping projects (patios, retaining walls, irrigation) valued up to \$100,000, meeting schedule and client specifications.
- Operated excavators and skid steers to prep sites and complete construction work, improving productivity, quality, and jobsite safety.
- Diagnosed and repaired mechanical issues on heavy and small equipment, applying hands-on engineering principles to maintain uptime and operational efficiency.

TECHNICAL SKILLS

Software: SolidWorks (3D modeling, assemblies, technical drawings), Fusion 360 (3D modeling, assemblies, CAM), AutoCAD (2D Drafting, Layouts), MATLAB, Project Management, Finite Element Analysis (FEA)

Machining & Fabrication: CNC Mills & Lathes, Manual Mills & Lathe